

# Esteban Naranjo

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## TECHNICAL SKILLS

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**Programming Languages:** SQL, Python, Java | Familiar with: MATLAB, R, Prolog

**Tools:** Neo4j, Git, LaTeX, Microsoft Office Suite

**Languages:** English (C1), French (B1), Dutch (A1), Spanish (native)

## EDUCATION

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**BSc, Data Science and Artificial Intelligence**

Maastricht, Netherlands

*Maastricht University*

2023 – 2027

- Relevant coursework: Data Analysis, Recommender Systems, Intelligent Systems, Databases, Software Engineering, Operations Research, Philosophy and AI

## EXPERIENCE

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**R&D Working Student | *Python, Geospatial data***

Sep. 2025 – Jan. 2026

*InnovatieLab Politie (via Maastricht University)*

Maastricht, Netherlands

- Developed a Flask web framework for visualizing a graph network using JSON data to handle and store location-based data
- Implemented a heuristic routing algorithm that uses area-based crime patterns for efficient police patrolling
- Performed data transformation, cleaning and loading of open crime data (geospatial data, time-based data) for full integration with the codebase

## PROJECTS

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**BSc Thesis: KG-Embeddings for Explainable Recommendation | *Python, Neo4j***

Feb. 2026 – Dec 2026

- Scraping pre-collected open datasets of restaurant and menu data
- Building a Knowledge Graph to implement Knowledge Graphs Embeddings and Similarity Scoring
- Extracting Path-based explanations from Knowledge Graphs using Neo4j and Cypher

**Handwritten Digits Recognition | *Python, OpenCV, Tensorflow***

March 2026 – May 2026

- Image recognition and processing of +17000 labelled handwritten Hindi numeral digits (0-9)
- Trained models using techniques Random Forests, Convolutional Neural Networks (CNN) for digits prediction with 99.86% accuracy on the validation set
- Placed Top 10 of the cohort on the leaderboard for this Kaggle competition

**Checkers AI Game Engine | *Java, javaFX, Maven, Git***

Sep 2024 – Feb 2025

- Developed a game engine for a checkers game using Java and JavaFX
- Implemented multi-agent playing environment using adversarial search techniques (Alpha-Beta search, Minimax)
- Implemented continuous delivery and persistent version control using Git and Github

**Pizza Delivery App | *SQL, MySQL, Java***

Sep 2024 – December 2024

- Developed a full-stack Java application for pizza delivery and stock management
- Implemented relational databases using MySQL for data storage and connection with the core application with MySQL Connector
- Implemented delivery drivers availability logic and scheduling to ensure a end-to-end pipeline, from ordering to checkout to delivery assignments

## CERTIFICATIONS

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**Neo4j Certified Professional**

August 2026

*Neo4j Graph Academy*